Lab: Executive Function -- How Your Prefrontal Cortex Plans the Future

Objective

Investigate the biological basis of **planning** through the prefrontal cortex, executive function, and goal-directed behavior, connecting brain development to the Active Inference concept of future-oriented simulation.

Prerequisites

- Completed all previous Biology & Health modules
- Read the module on Executive Function, Goal-Setting, and the Prefrontal Cortex

Part 1: Prefrontal Cortex Function Mapping

Goal: Map the specific planning functions of the prefrontal cortex.

Research the subdivisions of the prefrontal cortex and their roles:

Region	Location	Planning Function	Active Inference Role
Dorsolateral PFC		Working memory, sequencing	Policy evaluation
Ventromedial PFC		Value assessment, risk	Expected reward computation
Orbitofrontal cortex		Outcome prediction	Forward model simulation
Anterior cingulate cortex		Conflict monitoring	Prediction error detection

Write your response here

Part 2: Executive Function Self-Assessment

Goal: Assess your own executive function abilities and connect them to biology.

Try each task and rate your performance (1-5):

Executive Function	Task	Your Rating	Brain Region Involved
Working memory	Remember a 7-digit number for 30 seconds while counting backward		
Inhibition	Read color words printed in different-colored ink (Stroop test)		
Cognitive flexibility	Sort cards by one rule, then switch to a different rule		
Planning	Describe step-by-step how you would organize a class event		

Which tasks were hardest? Why might these be harder for teenagers than adults? (Connect to prefrontal cortex development.)

Write your response here

Part 3: Impulse Control and the Marshmallow Test

Goal: Understand the biology of delayed gratification.

Research the famous "marshmallow test" experiment and answer:

1. What were children asked to do?
2. What brain regions are involved in resisting temptation (waiting)?
3. What brain regions push for immediate reward?
4. In Active Inference terms, why is delayed gratification a planning problem? (Hint: it requires simulating future outcomes and valuing them over present sensations.)
5. Why is this particularly challenging for the teenage brain?

Write your response here

Part 4: Goal-Setting as Biological Planning

Goal: Design a personal goal using biological planning principles.

Choose a real goal you want to achieve in the next month. Map it biologically:

1. **Goal state (preferred future state)**: What does success look like?
2. **Current state**: Where are you now?
3. **Action sequence (policy)**: What steps will you take?
4. **Prediction errors to expect**: What obstacles might arise?
5. **Executive functions required**: Which prefrontal capacities will you need most?
6. **Health factors**: What sleep, exercise, or stress management will support your planning brain?

Write your response here

Summary Table

Concept	Brain Structure	Active Inference Connection
Executive function	Prefrontal cortex	Policy evaluation and selection
Working memory	Dorsolateral PFC	Maintaining states during planning
Impulse control	Ventromedial PFC vs. limbic system	Balancing present vs. future predictions
Goal-directed behavior	Orbitofrontal cortex	Simulating future outcomes